

Supplementary Material for “Pilot-Scale Text Column Detection and Reading-Order Recovery for Traditional Mongolian Historical Archives”

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This supplementary material provides diagnostic experiments, extended inter-annotator agreement per-page details, and transfer-detector baselines that supplement the main JKSU CIS manuscript.

1 Inter-Annotator Agreement — Per-Page Breakdown

Table 1 reports per-page agreement scores. The per-page breakdown reveals that Ignore-region agreement varies substantially by page (0.000–0.571), consistent with the observation that ambiguous artifact boundaries are the primary source of annotation disagreement. Text-column agreement is consistently high on pages with clear column boundaries ($F1 \geq 0.857$ for three of five pages), while the lowest box F1 (0.606 on page 80-48-73-1) reflects a page with irregular short columns and fragments that the two annotators interpreted differently.

2 Supplementary Transfer Baselines

Both RT-DETR-L and DocLayout-YOLO were fine-tuned for 5 epochs under the same train/validation/test split used for YOLOv8n. Because their training budgets are

Table 1 Per-page inter-annotator agreement metrics on the 5-page independent subset.

Page ID	GT cols	Box F1@0.5	Mean IoU	RO Agree	Ignore F1
80-48-70-2	22	1.000	0.836	0.909	0.000
80-48-62-1(1)	12	0.857	0.766	0.000	0.200
80-48-69-4	2	0.667	0.669	—	0.571
80-48-72-2	10	0.952	0.801	0.889	0.000
80-48-73-1	20	0.606	0.765	0.911	0.400
Aggregate	66	0.852	0.800	0.677	0.286

Table 2 Transfer-detector baselines on the original 11-page test split under the same validation-selected threshold protocol. These diagnostic results suggest that off-the-shelf modern document-layout priors transfer poorly to sparse vertical handwritten archive columns without stronger domain adaptation.

Method	Prec.	Rec.	F1	Mean IoU	RO Acc.
RT-DETR-L + rotation-aware order	0.032	0.270	0.058	0.632	0.847
DocLayout-YOLO + rotation-aware order	0.046	0.031	0.037	0.572	0.000

Table 3 Strict-localization comparison on the original 11-page test split at IoU@0.75. The stricter threshold penalizes boundary errors more heavily.

Method	Prec.	Rec.	F1	Mean IoU	RO Acc.
Heuristic column extraction	0.322	0.447	0.375	0.901	1.000
YOLOv8n + rotation-aware order	0.579	0.434	0.496	0.826	1.000
Faster R-CNN MobileNet + rotation-aware order	0.352	0.289	0.318	0.828	1.000

shorter than YOLOv8n (50 epochs) and their pretraining domains are dominated by modern printed document layouts, readers should interpret these results as diagnostic evidence of non-trivial domain transfer rather than as definitive model rankings. A fair-budget retraining comparison is left to future work.

3 Strict-Localization Comparison

4 Inference-Size Sensitivity

Table 4 reports inference-only sensitivity without retraining. Higher inference resolution improves recall at the cost of slower throughput. Note that these values are obtained by running the same 960-trained checkpoint at different test resolutions; full retraining at each image size is recommended before drawing training-configuration conclusions and is left to future work.

Table 4 YOLOv8n inference-only image-size sensitivity on the original 11-page test split. The model is trained at image size 960; inference is run at the listed sizes without retraining.

Inference size	Prec.	Rec.	F1	Mean IoU	RO Acc.
640	0.898	0.579	0.704	0.830	0.900
960	0.956	0.717	0.820	0.770	0.900
1280	0.965	0.737	0.836	0.753	0.900

Table 5 YOLOv8n retrained from scratch at three image sizes (50 epochs each) on the 43-page training set, evaluated on the expanded 54-page test set. Thresholds selected by validation F1.

Training image size	Prec.	Rec.	F1	Mean IoU	RO Acc.
640	0.898	0.852	0.875	0.768	0.997
960	0.902	0.875	0.888	0.770	0.997
1280	0.931	0.813	0.868	0.767	0.997

Table 6 YOLOv8n project-level test performance on the original 11-page test split with different numbers of annotated training pages.

Training pages	Test Prec.	Test Rec.	Test F1	Mean IoU	RO Acc.
7	0.843	0.388	0.532	0.719	1.000
20	0.875	0.691	0.772	0.748	0.900
43	0.956	0.717	0.820	0.770	0.900

5 Multi-Scale Training Ablation

Table 8 reports the effect of retraining YOLOv8n from scratch at three different image sizes with the same 50-epoch budget. Image size 960 achieves the best F1 (0.888), confirming the parameter choice used in the main manuscript. Image size 640 achieves identical F1 (0.875) with a lower threshold (0.20 vs. 0.15 for 960), while 1280 achieves the highest precision (0.931) but lower recall (0.813), consistent with the observation that higher-resolution training encourages more conservative detection. Unlike the inference-only sensitivity table (Table 4), these models were fully retrained at each image size, providing a more reliable picture of the resolution–performance trade-off.

6 Training-Set Size Scaling

7 Column-Aware Post-Processing Diagnostic

Table 6 evaluates a simple column-aware geometric filter that retains YOLO predictions satisfying validation-selected constraints on aspect ratio, relative height, and relative width. The column-aware filter improves validation F1 (0.854 vs. 0.792) but

Table 7 Diagnostic column-aware YOLO post-processing on the original 11-page test split.

Setting	Split	Prec.	Rec.	F1	Mean IoU	RO Acc.
Main YOLO threshold 0.35	test	0.956	0.717	0.820	0.770	0.900
Recall-oriented threshold 0.30	test	0.937	0.776	0.849	0.761	0.818
Column-aware prior filter	val	0.929	0.791	0.854	0.779	0.875
Column-aware prior filter	test	0.949	0.730	0.825	0.760	0.818

Table 8 Ablation of ignore-region filtering and rotation-aware reading order at score threshold 0.35 on the original 11-page test split.

Configuration	Prec.	Rec.	F1	RO Acc.	FP/FN	Removed
Full: ignore + rotation-aware order	0.956	0.717	0.820	0.900	5 / 43	0
w/o rotation-aware order	0.956	0.717	0.820	1.000	5 / 43	0
w/o ignore filtering	0.956	0.717	0.820	0.900	5 / 43	0
w/o both	0.956	0.717	0.820	1.000	5 / 43	0

does not robustly dominate the main threshold on the test split (F1 0.825 vs. 0.820) and reduces reading-order accuracy (0.818 vs. 0.900). This trade-off, combined with the risk of overfitting geometric priors to the small validation set, motivates keeping the simple threshold as the main result.

8 Ablation of Ignore-Region Filtering and Rotation-Aware Ordering

Table 7 ablates two post-processing components on the original 11-page test split. Ignore-region filtering does not suppress any YOLO predictions on this test set (no prediction overlaps an Ignore region by ≥ 0.3 IoU), so detection F1 is unchanged. Rotation-aware ordering also leaves detection F1 unchanged; on this split, the simple x-sorted order happens to give slightly higher reading-order accuracy than the rotation-aware heuristic, reinforcing that the ordering rule is a transparent proxy rather than a fully solved component.

9 Runtime Comparison

Table 8 reports per-page inference time on the original 11-page test split measured on a local CPU (Mac M5 Pro). The rule-based pipeline is the slowest (1.84 s/page). The two learned detectors are faster and close to each other (0.17–0.19 s/page); these timings are implementation-specific and should not be interpreted as general speed rankings, because batching, preprocessing, and hardware can change the relative ordering.

Table 9 Local CPU runtime on the original 11-page test split.

Method	Device	Mean s/page	Median s/page
Heuristic column extraction	CPU	1.843	1.729
YOLOv8n + rotation-aware order	CPU	0.193	0.196
Faster R-CNN MobileNet + rotation-aware order	CPU	0.168	0.162